
Abstraction

An overview of the abstract side of mathematics, to put us in a good state of mind for category theory. It's important not to be thinking of math in the common narrow way as numbers, equations and problem solving. This chapter will still have little formality.

2.1 What is math?

In the previous chapter I described category theory as “the mathematics of mathematics”. So I’d better start by describing in more depth how I think of mathematics. If you take too narrow a view of what math is, then the phrase “the mathematics of mathematics” doesn’t make any sense.

Math is not just the study of numbers and equations, and it’s not all about solving problems. Those are some aspects of math, and they are often the aspects that are emphasized in school math, and in math for non-mathematicians. Math in school tends to start with numbers and arithmetic, moves on to equations, and then maybe deals with things like trigonometry and a bit of geometry. Trigonometry and geometry aren’t really about numbers but about shapes; however they still involve a lot of numbers and things like calculating angles and lengths, expressed in numbers.

Math as used in the world does involve quite a bit of solving problems using numbers and equations. There are usually some measurable quantities measured in numbers, and some equations relating them, and the task is to calculate the ones we don’t know, using the ones we can measure.

At least, this is the most visible and obvious way in which math is used in the world, so I don’t blame anyone for thinking that’s all there is. But there is more going on behind that: the *theory* of how those things work. That theory is what enables us to make sure they work, to refine them, and to develop new versions to deal with more complex and nuanced situations. This is like the foundations of a building: you can’t see them, but without them the building would not stand up.

Every academic discipline provides a way of reaching truths of some form. Each discipline is seeking a particular type of truth, and develops a method or framework for deciding what counts as true. In this era of information excess (and indeed general excess) I think understanding those methods and frameworks is far more important than knowing the truths themselves. The important thing is to know *how* to decide what should count as true — how to build good foundations on which to base our understanding. I strongly believe that this understanding of process and framework is what is most transferrable about studying any subject, especially math.

2.2 The twin disciplines of logic and abstraction

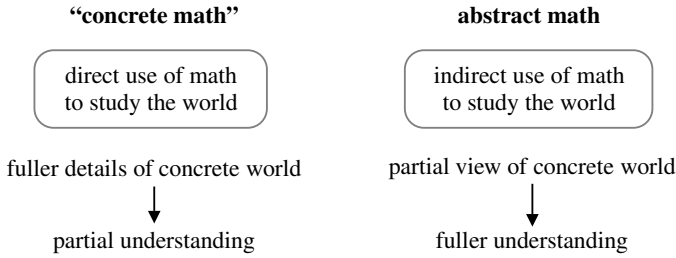
The framework of mathematics involves the twin disciplines of logic and abstraction. Math is not unique in its use of either of these things, but I regard it as being more or less defined by its use of these in combination.

I would say that philosophy uses logic, but applies it to real questions about life experiences. Art uses abstraction, but does not primarily build on its abstractions by logic. Math uses logic and abstraction together. It uses logic to build rigorous arguments, and uses abstraction to ensure that we are working in a world where logic can be made rigorous.

This might make it sound like we can never be talking about the “real” (or rather, concrete[†]) world as we will always be working in an abstract world. While this is in some sense true, it is also reductive. Abstractions are facets of the concrete world, or views from a particular angle. While they will never give us the full explanation of the concrete world, it is still valuable to get a very full understanding of particular aspects of the concrete world. As long as we are clear that each one is only a partial view, we can then move flexibly between those different views to build up a clearer picture.

There is a subtle difference between this and the approach of studying the concrete world directly. In the direct approach we typically get only a partial understanding, because the concrete world is too messy for logic. The following diagram illustrates the difference.

[†] What is real anyway?



Thus abstract math still studies the world, just in a less direct way. Its starting point is abstraction, and the starting point for abstraction is to forget some details about a situation.

2.3 Forgetting details

Abstraction is about digging deep into a situation to find out what is at its core making it tick. Another way to think of it is about stripping away irrelevant details, or rather, stripping away details that are irrelevant to what we’re thinking about now. Those details might well be relevant to something else, but we decide we don’t need to think about them for the time being. Crucially, it’s a careful and controlled forgetting of details, not a slapdash ignoring of details out of laziness or a desire to skew an argument in a certain direction.

If someone says “women are worse at math than men” then they are omitting crucial details and opening up ambiguities, or deliberately using data in a misleading way. This inflammatory statement has some truth *in some sense*, which is that fewer women are currently employed as math professors than men, not that there is any evidence that women are *innately* worse at math than men. It’s a pedantically correct expression of the fact that women are currently doing worse in the field of mathematics than men are.[†]

Whereas if we observe that one apple together with another one apple makes two, and that one banana together with another banana makes two, and we say that one thing together with another thing makes two, then we are ignoring the detail of applehood and bananahood as that is genuinely irrelevant to the idea of one thing and another thing making two things. That is abstraction, and is how numbers come into being. Numbers are one of the first abstract concepts we come across, but we don’t always think of them as being abstract. That is a good thing, as it shows that we have raised the baseline level of abstraction that we’re comfortable with in our heads. This is like the fact that things can

[†] In *The Art of Logic* I wrote about pedantry being precision without illumination. In this case it’s even worse: it’s precision with active obfuscation.

seem hard at first, but later seem so easy they're second nature. It is all a sign that we've progressed.

2.4 Pros and cons

Before I go into more detail about how we perform abstraction, I want to talk about the pros and cons. It might be tempting just to talk about all the benefits of doing something, but that can be counter-productive if other people see disadvantages and think you're being dishonest or misleading. Instead, I think it's important to see the pros and cons of doing something, and weigh them up. There are rarely exclusively positives or negatives to doing something.

The advantages of abstraction, as I see them, are broadly that we unify many different situations in a sort of inclusivity of examples; this then enables us to transfer insight across different situations, and thus to gain efficiency in our thought processes by studying many things at once. The world is a complicated place and we need to simplify it in order to be able to understand it with our poor little brains. One popular way to simplify it is to ignore some of the detail, but I think that's a dangerous way of simplifying it. Another way is to make connections inside it so that our brains can deal with more of it at once. I think this is a better way. The best way overall is to become more intelligent so that the world becomes simpler relative to our brains.

So much for the advantages; I will now acknowledge some disadvantages of abstraction. One is that it does take some effort, but I do think this is about front-loading effort in order to reap rewards from it later. I think that it's an investment, and the extra effort *early* means that we can understand more things more deeply with less effort *later*.

Another disadvantage is that we lose details. However, I think this just means we shouldn't remain *exclusively* in the abstract world, but should always bear in mind that at some point some details will need to go back in. Losing the details temporarily is an important part of finding connections between situations, so again I think that this is a net positive.

Another disadvantage is that this takes us further from the normal everyday world that we're used to, which can be scary. It can be scary because it can seem like we don't have our feet on the ground any more. It can be scary because we can't touch things or feel things or see things, and we can't use a lifetime of intuition any more. However, intuition is itself a double-edged sword.[†] Intuition both helps and hinders us, whether in math or in life. It helps

[†] I've always found the metaphor of a double-edged sword a bit strange, as it doesn't seem to me that the two edges of a double-edged sword work in opposition to one another.

us in situations where we do not have enough information or time to use logic. It helps us by drawing on our experience quickly. But it is thus also limited by our experience, and if it is used instead of logic it can be dangerously misleading. For example, it's unavoidable to have a gut instinct when we meet a new person, but it's wrong to hold onto that instead of actually responding to the person as we get to know them, especially when our gut instinct is skewed by implicit bias as it (by definition) always is.

Likewise, in math it's not wrong to have intuitions about things, and indeed this is how much research gets started, by a vague idea coming from inside a mathematician's figurative gut somewhere. But the key is then to investigate it using logic and not rely too much on that intuition.

One crucial point is that the framework of building arguments by rigorous logic in math can take us much further than our intuition can. It can take us into places where we have no intuition, such as infinite-dimensional space, or worlds of numbers that have no concrete interpretation in the normal world. For example, one of the points of calculus is to understand what gives rise to interesting features in graphs, like gaps, spikes, places where the graph changes direction. This means we can seek those features even when the graph itself is much too complicated to draw, so that we can't just look for them visually.

One possible objection to this "advantage" is that you might think you'll never find yourself in a place that's so far beyond your intuition. That may well be true. But it might still be good for your brain to be stretched into those places, so that your intuition can develop. I am convinced that my years of stretching my brain in those abstract ways have enabled me to think more clearly about the world around me, and more easily make connections between situations, connections that others don't see. Often when I give my abstract explanations of arguments around social issues such as sexual harassment, sexism, racism, privilege, power imbalance and so on, people ask me how I thought of it. The answer is that a lifetime of developing my abstract mathematical brain makes these things come to me quite smoothly. It's good to train yourself to be able to do more than you think you'll need, so that the things you do need to do feel easier.

The last advantage I want to give for abstraction is that it can be fun. Fun can seem a little frivolous in trying political times, but if we only stress the utility of something it might start sounding awfully boring. I find it enormously satisfying to strip away outer layers of a situation to find its core. It appeals to my general aesthetic, which is typically that I am not so interested in superficial appearances, but care about what is going on in the heart of things, deep down, far below the surface. Abstraction in its own right really does bring me joy.

2.5 Making analogies into actual things

Abstraction comes from seeing analogies between different situations. This is a particular form of detail-forgetting, based in finding connections between different situations, rather than just arbitrarily ignoring details.

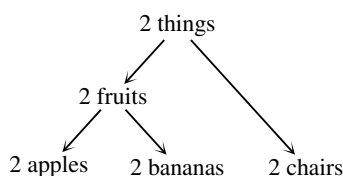
If we say that one situation “is analogous to” another situation, essentially what we are saying is that if we ignore some surface-level details in each situation then the two are really the same. In math, unlike in normal life, we don’t just say that the situations are “analogous”, but we make very precise what feature is the same in both situations, which is causing the analogy we want to consider. Some of what follows I have also written about in *The Art of Logic*.

If we think about two apples and two bananas, we can consider them to be analogous because they’re both examples of two things. But we could also consider them to be analogous because they’re both examples of two fruits. Neither of those is “right” or “wrong”, neither is “better” or “worse”. What we can say, however, is that “two things” is a further level of abstraction than “two fruits” because it forgets more of the details of the situation; conversely, saying “two fruits” is less abstract and leaves us closer to the actual situation.

The more abstract version takes us further away from “reality” (whatever that is), and one major upside is that it enables us to include more distant examples in our analogy. In this example it means we could include two chairs, or two monkeys, or two planets.

In a way, abstraction is like looking deeper into a situation, but it is also like taking a step back and seeing more of the big picture rather than getting lost in the details. The fact that we can find different abstractions of the same thing makes it sound ambiguous but is in fact key. I find it helpful to draw diagrams of different levels of abstraction.

Here is a diagram showing that two apples and two bananas are analogous because they’re both two fruits, but also that if we go up further to the level of two things then we get to include two chairs as well.



The key in math is that we don’t just say that things are *analogous*. Rather, we precisely specify our level of analogy, and then go a step further and regard that as an object in its own right, and study it. That is how we move into abstract worlds, and it is in those abstract worlds that we “do” math. In the above example that’s the level of “2 things”: the world of numbers.

Pinning down what is causing the analogy, rather than just saying things are analogous, is like the difference between telling someone there is a path

through a field, and actually marking out where the path is. I remember going on hiking expeditions in the rolling hills of the South Downs in the south-east of England when I was growing up. We would climb over a stile[†] into a field, and a little wooden signpost would tell us that a path was that way. But it didn't actually tell us where the path was, and invariably we would get to the other side of the field and no stile would be in sight because we had strayed from the supposed "path".

Pinning down structure rather than just observing that it exists is an idea that we will come back to, especially when we are doing more nuanced category theory later on. But for now the main point is that pinning things down helps us iron out some ambiguity, especially as there is always the possibility of different abstractions of the same situation.

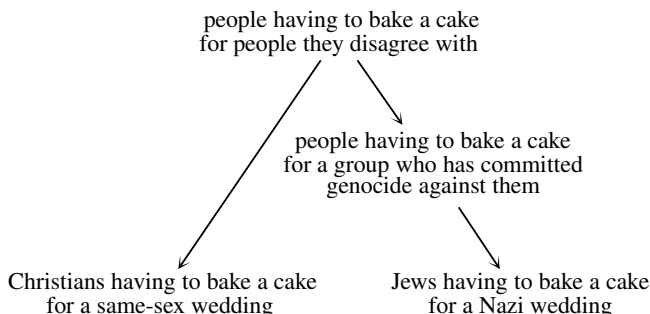
2.6 Different abstractions of the same thing

It might seem confusing that there can be different abstractions of the same thing, but in a way that's the whole point. Abstraction is not a well-defined process, which is to say that it can produce different results depending on how you do it. (In math "well-defined" means "unambiguously determined"; it doesn't just vaguely mean that a definition has been done well.) This is the cause of many acrimonious arguments in normal life, because if we just declare something to be "analogous to" something else, it leaves open an ambiguity of what level we're talking about. Then someone else is likely to get angry and either say "It's not the same, because. . ." and then point out some way in which your two things are different, or they will say "Well that's the same as saying. . ." and then invoke some ludicrous level of abstraction and implicitly accuse you of using that level. In the above example the first case would be like someone saying "2 apples aren't the same as 2 bananas because you can't eat the skin of bananas", and the second case would be like someone saying "Well if you think 2 apples are the same as 2 chairs you've obviously never tried sitting on an apple!"

Here is a much less trivial example. In 2018 in Colorado there was a court case involving some Christian bakers who refused to bake a cake for a same-sex wedding. I saw someone online (no less) who tried to argue that saying Christians should have to bake a cake for a same-sex wedding is like saying Jews should have to bake a cake for a Nazi wedding. I sincerely hope that your gut

[†] A stile is a traditional form of wooden step to enable people to climb over a fence into or out of a field, without letting livestock out. They're quite common in the UK where there are often "rights of way" giving the public the legal right to walk through a field.

reaction is “That’s not the same!” but I actually think it’s important to be able to acknowledge the *sense in which* those things are analogous, even if this sense disgusts us. I really think that just retorting “That’s not the same” is ineffective, and that using some more careful logic is more productive. Here is a level of abstraction that does yield that analogy, and a different level that differentiates between those two conclusions:



My wonderful PhD supervisor, Martin Hyland, taught me that when you’re doing abstract mathematics, the aim shouldn’t be to find the most abstract possible level, but to find a good level of abstraction for what you’re trying to do. “Horses for courses” one might say.[†] He also instilled in me the idea of starting sentences with “There is a sense in which. . .” because math isn’t about right and wrong, it’s not about absolute truth; it’s about different contexts in which different things can be true, and about different senses in which different things can be valid. Abstraction in mathematics is about making precise which sense we mean, so that instead of having divisive arguments, whether it’s about abstract theories or about homophobic bakers, we can investigate more effectively what is causing certain outcomes to arise.

In the end, we could always make absolutely anything the same by forgetting just about all of the details, like when I take my glasses off and everything looks equally blurry. We do this in life as well: sometimes it is tempting to declare that we’re all human, so we’re all the same really. This might sound happily unifying, but it also negates various people’s struggles against oppression, prejudice, poverty, illness and any number of other things. There is the opposite tendency as well, to fragment into so many different identities and combinations of identities, to emphasize our unique experiences. Pushing too far to either extreme is probably unhelpful. What is helpful is to be able to see what all the levels are, and maintain enough flexibility to be able to move between them and draw different insights from different levels.

[†] One also might not, if one has never heard of this saying.

Things To Think About

T 2.1 What are some senses in which addition and multiplication are “the same”? What are some senses in which they are “different”?

Addition and multiplication are both *binary operations*: they take two inputs and produce one answer at the end. In the first instance they are binary operations on numbers, but as math progresses through different levels of abstraction we find ways of defining things like addition and multiplication on other types of mathematical object as well.

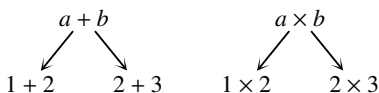
As binary operations, they have some features in common, including that the order in which we put the numbers doesn’t matter (which is *commutativity*) and nor does how we parenthesize (which is *associativity*). However, addition and multiplication behave differently in various ways. For example, addition can always be “undone” by subtraction, but multiplication can only sometimes be “undone” by division: multiplication by 0 can’t be undone by division. This is sometimes thought of as “you can’t divide by 0” but we’ll come up with some better abstract accounts of this later.

2.7 Abstraction journey through levels of math

As mathematics progresses, aspects of it become more and more abstract. There is a sort of progression where we move through levels of abstraction gradually, through the following steps:

1. see an analogy between some different things,
2. specify what we are regarding as causing the analogy,
3. regard that thing as a new, more abstract, concept in its own right,
4. become comfortable with those new abstract concepts and not really think of them as being that abstract any more,
5. see an analogy between some of those new concepts,
6. iterate...

One of the advantages of taking abstract concepts seriously as new objects is that we can then build on them in this way. Here’s an example of an initial process of abstraction in basic math.



This is the infamous process of “turning numbers into letters”, which is the stage of math many people tell me is where they hit their limit.

(In fact there was a level below that, where we went from objects like apples and bananas to numbers in the first place.) Why have the numbers turned into letters? It's so that we can see things that are true abstractly, regardless of what exact numbers we're using. For example:

$1 + 2 = 2 + 1$	Something analogous is going on in all of these situations, and it would be impossible for us to list all the combinations of numbers for which this is true as there are infinitely many of them. We could describe this in words as "if we add two numbers together it doesn't matter what order we put them", but this is a bit long-winded.
$1 + 3 = 3 + 1$	
$2 + 3 = 3 + 2$	
$5 + 7 = 7 + 5$	
$\vdots$	

The concise abstract way of saying it is: given any numbers a and b ,

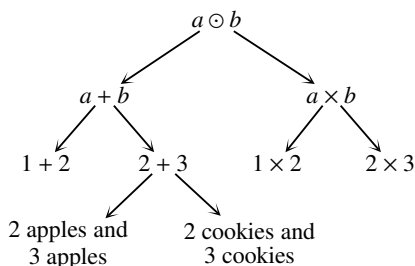
$$a + b = b + a.$$

We have "turned the numbers into letters" so that we can make a statement about tons of different numbers at once, and make precise what pattern it is that is causing the analogy that we see. Not only is this more concise and thus quicker to write down (and mathematicians are very lazy about writing down long-winded things repeatedly), but the abstract formulation can help us to go a step further and pin down similarities with other situations.

But there's a level more abstract as well, in the direction we were going at the end of the previous section. If we think about similarities between addition and multiplication we see that they have some things in common. For a start, they are both processes that take two numbers (at a basic level) and use them to produce an answer. The processes also have some properties that we noticed, such as commutativity and associativity.

When we call them a "process that takes two numbers and produces an answer" that is a further level of abstraction. It's an analogy between addition and multiplication.

Here is a diagram showing that new level, with the symbol $\odot$ representing a binary operation that could be $+$, $\times$ or something else. There is a journey of abstraction up through levels of this diagram that is a bit like the journey through math education.

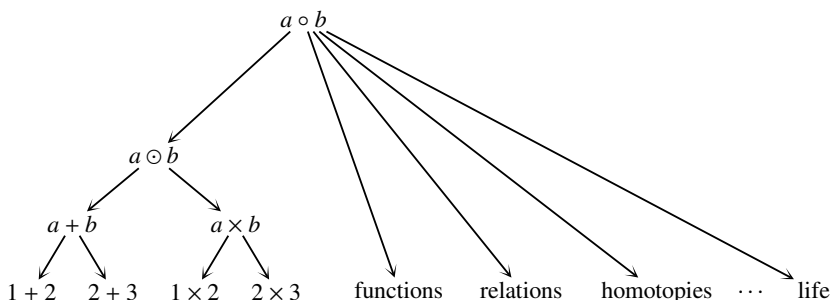


At the very bottom level we have the sorts of things you might do in pre-school or kindergarten where you play around with familiar objects and get nudged

in the direction of thinking about numbers. At the next level up we have arithmetic as done in elementary school, perhaps, and then we move into algebra as done a little bit further on at school. The top level here, with the abstract binary operation, is the kind of thing we study in “abstract algebra” if we take some higher level math at university. Binary operations are studied in group theory, for example, and this is one of the topics we’ll come back to. Incidentally I always find the term “abstract algebra” quite strange, because all algebra is abstract and, as I’ve described, what we even consider to count as “abstract” changes as we get more used to more abstraction.

There is indeed a further level of abstraction, which one might call “very abstract algebra”. At this level we can think about more subtle ways of combining two things, where instead of taking just any two things and producing an answer, we can only take two things that fit together like in a jigsaw puzzle. For example, we can think about train journeys, where we can take one train journey and then another, to make a longer train journey — but this only makes sense if the second journey starts where the first one ends, so that you can actually change train there. This means you can’t combine *any old* train journeys to make longer ones, but only those that meet up suitably where one ends and the other begins.

This is the sort of way we’ll be combining things in category theory. Binary operations are still an example of this, but as with all our higher levels of abstraction, we will now be able to include many more examples of things that are more subtle than binary operations. This includes almost every branch of math, as they almost all (or maybe even all) involve some form of this way of combining things. Here is a diagram showing that, including the names of some of the mathematical topics we’ll be exploring later in this book.



I’ve included “life” in the examples here, to emphasize that the higher level of abstraction may seem further away from normal life, but at the same time, the higher level is what enables us to unify a much wider range of examples, including examples from normal life that are not usually included in abstract

mathematics. I think this is like swinging from a rope — if you hang the rope from a higher pivot then you can swing much further, provided you have a long rope. It's also like shining a light from above — if you raise it higher then the light will become less bright but you will illuminate a wider area. The result is perhaps more of an overview and an idea of context, rather than a close-up of the details. However, you won't lose the details forever as long as you retain the ability to move the light up and down. Furthermore, if you can find a way to make the light itself brighter, then you can see more detail and more context at the same time. I think this is an important aspect of becoming more intelligent, and that abstract mathematics can help us with that.

“More abstract” doesn't necessarily mean less relevant and it doesn't necessarily mean harder either. Too often we assume that things get harder and harder as we move up through those levels of abstraction, and thus that we shouldn't try to move up until we've mastered the previous level. However I think this is one of the things that can hold some people back or keep them excluded from mathematics. Actually the higher levels might be easier for some people, either because, like me, they enjoy abstraction and find it more satisfying, or because it encompasses more examples and those might be more motivating than the examples included at the lower levels. If you're stuck at the level of $a + b$ and the only examples involve adding numbers together then you may well feel that the whole thing is tedious and not much help to you either. After all, some things are fun, some things are useful, and some things are both, but the things that are neither are really the pits.

I think we should stop using the lower levels of abstraction as a prerequisite for the higher ones. If at the higher levels you get to deal with examples from life that you care about more than numbers, perhaps examples involving people and humanity, then it could be a whole lot more motivating. Plus, if you enjoy making connections, seeing through superficiality, and shining light, then those higher levels are not just useful but also fun.